indirect arguments for their existence we've yet to say anything about what these constituents might actually be. And for good reason. We really have no idea what they are. Or, perhaps I should say, when it comes to identifying spacetime's elemental ingredients, we have no ideas about which we're really confident. This is a major gap in our understanding, but it's worthwhile to see the problem in its historical context.

Were you to have polled scientists in the late nineteenth century about their views on matter's elementary constituents, you wouldn't have found universal agreement. A mere century ago, the atomic hypothesis was controversial; there were well-known scientists—Ernst Mach was one—who thought it wrong. Moreover, ever since the atomic hypothesis received widespread acceptance in the early part of the twentieth century, scientists have been continuously updating the picture it supplies with what are believed to be ever more elementary ingredients (for example, first protons and neutrons, then quarks). String theory is the latest step along this path, but because it has yet to be confirmed experimentally (and even if it were, that wouldn't preclude the existence of a yet more refined theory awaiting development), we must forthrightly acknowledge that the search for nature's basic material constituents continues.

The incorporation of space and time into a modern scientific context goes back to Newton in the 1600s, but serious thought regarding their microscopic makeup required the twentieth-century discoveries of general relativity and quantum mechanics. Thus, on historical time scales, we've really only just begun to analyze spacetime, so the lack of a definitive proposal for its "atoms"—spacetime's most elementary constituents— is not a black mark on the subject. Far from it. That we've gotten as far as we have—that we've revealed numerous features of space and time vastly beyond common experience—attests to progress unfathomable a century ago. The search for the most fundamental of nature's ingredients, whether of matter or of spacetime, is a formidable challenge that will likely occupy us for some time to come.

For spacetime, there are currently two promising directions in the search for elementary constituents. One proposal comes from string theory and the other from a theory known as *loop quantum gravity.*